

AI-Assisted Marking and Assessment Analytics.

The highest value part is not “creating tests”. That’s already becoming commoditised. AI can generate tests easily now.

The real pain is:

- marking at scale
- moderation
- analytics
- audit trails
- reducing admin
- standardisation
- preventing disputes
- handling thousands of scripts quickly

That’s where institutions pay serious money. Schools and universities don’t lose sleep over “how do we make a test”.

They lose sleep over:

- turnaround times
- inconsistent marking
- missing marks
- spreadsheet chaos
- moderation disputes
- overloaded lecturers
- exam periods becoming operational disasters

That is the business.

TARGET MARKETS

Weak Market - Small Schools

- low budgets
- slow procurement

- lots of support needed
- resistance to change

Better Market - Universities & Colleges

- huge marking load
- pressure for faster turnaround
- digital transformation budgets
- lecturers drowning in admin

Potentially Best Market - Centralized Marking Centres & Exam Bodies

- thousands of papers
- temporary markers
- moderation workflows
- quality control problems
- inconsistent scoring
- spreadsheet hell
- physical logistics

RAG Usage

We only use it if:

- the system must reference institutional material
- past papers
- rubrics
- curriculum standards
- lecturer notes
- marking memos
- policy documents

E.g.

A university uploads previous exams, marking rubrics, course outcomes, and bloom taxonomy requirements. Then the AI generates aligned assessments, memo suggestions, marking guidance. This ensures no generic AI content forms around it.

Positioning

We must not position it as an AI system that marks everything automatically, but rather AI-assisted marking with human moderation.

The system:

- marks
- rubric alignment
- detected mistakes
- inconsistencies
- summaries

Human approves.

MVP

The MVP is not:

- an LMS
- Google Classroom competitor
- student portal
- full university ERP
- exam delivery platform

The MVP is: assessment marking and analytics infrastructure. Everything must support that core workflow. If a feature does not improve:

- marking speed
- marking quality
- reporting
- moderation

- operational efficiency

...then it probably doesn't belong in the first phase.

CORE USER ROLES

1. Institution Admin - (faculty admin, dean office, exam office, department operations).

- They control departments, lecturer accounts, permissions, billing, & institutional analytics.

2. Lecturer/Examiner - (main operational user).

- They upload assessments, upload rubrics, upload submissions, review AI suggestions, finalize marks, & release results.

3. Marker/Teaching Assistant - (limited permissions)

- They can mark assigned scripts, review AI suggestions, & leave comments.

- They cannot access institution-wide data, change rubrics, export sensitive reports.

4. Moderator - (critical role).

- The check for marking consistency, suspicious grading, outliers, & rubric adherence.

USER FLOW

1. Registration/Institution Setup

- Institution creates university profile, faculty, departments, & grading systems.

- E.g. Engineering faculty, computer science department.

- Admin invites lecturers, markers, and moderators.

- No social signups, enterprise software works best with invitation links, institutional emails, & role assignment.

2. Create Assessment

- Lecturer creates course, assessment, due date, assessment type.

- E.g. midterm, final exam, quiz, assignment, or lab report.

3. Upload Assessment Materials

- Question Paper + Rubric/Memo (Critical Uploads).
- Student submissions - pdfs, docx, or scanned papers. A smarter move would be to start with typed submissions and not handwritten ones unless OCR quality is excellent.

4. AI Processing Pipeline

- Step 1: Parse assessment - system must understand questions, sections, marks allocation, and expected answers.
- Step 2: Parse rubric - system extracts marking criteria, scoring rules, penalties, and weighing.
- Step 3: Evaluate submissions - AI generates suggested marks, reasoning, confidence score, rubric alignment. And it must never hide the reasoning - lecturers need explainability.

5. Human Review Interface

- Lecturer can see a table with: question, student answer, ai suggestion, rubric match, final mark.
- They can accept, modify, override, and/or leave comments.

6. Finalization

- Once approved marks locked, audit trail stored, and exports generated.

Exports

- CSV, Excel, & PDF summaries are essential. Universities still live on spreadsheets.

Analytics Categories

1. Student-Level Analytics

- Per student: overall score, section scores, weak concepts, strengths, rubric performance, historical trend. E.g. student struggles with database normalization questions.

2. Class-Level Analytics

- Class dashboard: average mark, median, pass rate, failure rate, distribution curve, hardest questions, easiest questions.
- Question-level analytics are gold. E.g. 78% of students failed Question 4.
- This signals bad teaching, ambiguous question, curriculum gap.

3. Department-Level Analytics

- Example: module performance trends, lecturers' comparison, assessment difficulty consistency, pass rate trends across semesters. Management will love this.

4. Institutional Analytics

- Example: turnaround time per department, marking workload, moderation disputes, grading consistency, & throughput analytics.

5. Marker Performance Analytics

- Example: marking speed, consistency, deviation from rubric, discrepancy rates, & moderation corrections.

Weak Area Detection

- The system can detect commonly failed concepts, recurring mistakes, and low scoring outcomes.

AI Confidence Scoring

- Each mark suggestion should have: confidence %, explanation, & highlighted uncertainty
- Low confidence answers should be flagged for human review thus protecting trust.

Audit Trails

- Every mark change must store: who changed it, when, original mark, new mark, and reason.
- Because disputes happen. And ensures traceability.

FINANCIAL MODEL

Pricing Structure: Per institution base fee + Usage-based pricing (price per script)

Pricing per script works because value scales with workload and institutions already understand printing costs, marking costs, & operational costs.

For big institutions like universities, we can have enterprise contracts, annual licensing, dedicated support, and/or integrations.

ROI for institutions

These institutions don't actually pay for AI. They pay for reduced turnaround time, fewer admin staff hours, reduced moderation load, consistency, reporting, operational visibility. That's their ROI.

Cost Drivers

AI Inference Cost - LLMs are expensive at scale, especially with long PDFs, thousands of scripts, and complex reasoning. We need batching, smaller specialized models, hybrid systems, and structured grading pipelines.